Navid Dejbord

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Research Profile

My research interests are in mathematical logic, particularly proof theory and provability logic. My previous work includes the provability logic of Peano Arithmetic and the (Σ_1)-provability logic of Heyting Arithmetic. More recently, I have been independently studying type theory and homotopy type theory, with the aim of developing further research in these areas. I am also open to related directions in logic, foundations of mathematics, and theoretical computer science.

Education

– **Master of Mathematics** *Mathematical Logic*
Sharif University of Technology, Tehran, Iran Graduated: September 2024
Relevant Coursework: *Constructive Mathematics, Proof Theory, Model Theory, Type Theory.*

– **Bachelor of Mathematics** *Pure Mathematics*
Sharif University of Technology, Tehran, Iran Graduated: September 2021
Relevant Coursework: *Foundations of Mathematics, Mathematical Logic, Modal Logic, Intuitionistic Logic, Gödel's Incompleteness Theorems.*

Research Projects

– **Σ_1-Provability Logic of Heyting Arithmetic** *(Masters's Thesis)* 2023–2024
Supervised by Mohammad Ardeshir, Advised by Mojtaba Mojtahedi
Studied the characterization of the Σ_1-provability logic of Heyting arithmetic, including its axiomatization, soundness, completeness, and decidability. Examined the role of realizability, NNIL/TNNIL approximation algorithms, and Solovay-style Kripke-model constructions in the completeness proof.

– **Provability Logic of Peano Arithmetic** *(Bachelors's Thesis)* 2020–2021
Supervised by Mohammad Ardeshir
Studied the formalization of provability in primitive recursive arithmetic, including Gödel numbering, the diagonal lemma, and Löb's theorem. Examined Gödel–Löb logic and the arithmetical completeness results of Solovay and Visser, including the case of Σ_1 substitutions.

Articles and Presentations

– **Decidability of Euclidean Geometry** 2025
Scientific Seminar Series for Students, Allameh Helli 5 High School.

– **Where Do You Get Your Crazy Ideas?** (translation to Persian) 2025
Ian Stewart, *Sharif Mathematics Journal*, 3(3) (to be published).

– **Transcendence of Numbers** 2024
Sharif Mathematics Journal, 3(2).

– **Proof Complexity** 2023
Computer Sciences Seminar, Sharif University of Technology, Department of Mathematical Sciences.

– **NNIL Algorithm** 2022
Logic Seminar, Sharif University of Technology, Department of Mathematical Sciences.

Academic Achievements

– **University Entrance Exam (Master's):** Rank 15 (Nationwide) 2021
Ranked among the top 1% *candidates in the nationwide master's entrance exam.*

– **University Entrance Exam (Bachelor's):** Rank 110 (Nationwide) 2016
Ranked among the top 0.1% *candidates in the nationwide master's entrance exam.*

Experiences

– **Mathematics Teacher** *High School Geometry*
Allameh Helli High School 2024–2026

– **Mathematics Editor**
Monta Educational Institution, Maze Educational Institution 2024–2026

– **Teaching Assistant** *Mathematical Logic, Logic for Computer Sciences*
Sharif University of Technology 2019–2025

– **Part-time Bookseller**
Sari Bookcity, Bookshaar Bookstore 2021–2024

Technical Skills

– Python, LaTeX, Lean, Agda

Current Status

– **Mandatory National Service** 2026–Present
Islamic Republic of Iran Army (Artesh)

Volunteer Activities

– **Editor-in-Chief** of *Sharif Mathematics Journal* 2023–2024
A student journal conducted by current and former students of the mathematics department at Sharif University of Technology.

– **Member** of *Yarigaran Center* 2018–2019
A student association at Sharif University of Technology dedicated to promoting educational justice by supporting disadvantaged students.